

YAZAN MAHER HAMDAN

Technical Lead · AI/ML Engineer · ERP Specialist · Full-Stack Developer
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PROFESSIONAL SUMMARY

Results-driven Technical Lead with expertise in AI engineering, enterprise ERP implementation, and full-stack software development. Architected and deployed RYAI — a production-grade, agentic AI platform natively integrated into ERPNext — encompassing Retrieval-Augmented Generation (RAG), multi-LLM orchestration, and intelligent document processing. Delivered end-to-end products across cybersecurity (SentriX SOC platform) and web infrastructure (drag-and-drop CMS). Published researcher in machine learning and network security with a consistent track record of leading cross-functional teams and translating complex business requirements into scalable, enterprise-grade solutions.

Core competencies span the full delivery lifecycle: system architecture, AI/ML model development, ERPNext/Frappe customisation, full-stack engineering (Python, React, FastAPI), team leadership, and client-facing technical consulting across financial services, HR, and industrial sectors.

CORE COMPETENCIES

AI & Machine Learning · Retrieval-Augmented Generation (RAG) · Large Language Models (LLM) · Deep Learning · NLP
ERPNext / Frappe Framework · ERP Implementation · Business Process Automation · Digital Transformation
Full-Stack Development · REST API Design · System Architecture · Cloud-Native Applications · Docker
Technical Leadership · Agile Project Management · Stakeholder Management · Team Mentoring · Client Consulting
Python · JavaScript / TypeScript · React · FastAPI · Django · SQL · Power BI · TensorFlow · Scikit-learn

PROFESSIONAL EXPERIENCE

Technical Lead | iValue Management Consulting

Jan 2025 – Present

- Spearheaded the architecture and end-to-end delivery of RYAI, a production-ready agentic AI platform embedded natively within ERPNext, featuring an OpenAI tool-use RAG pipeline, multi-LLM support (OpenAI, Gemini, Ollama), semantic vector search, CV auto-extraction, and SPA-compatible field-level AI injection
- Led and mentored a cross-functional team of developers and IT specialists across concurrent ERPNext implementation engagements, ensuring delivery against technical quality standards and project milestones
- Defined system architecture and translated complex enterprise business requirements into scalable, maintainable technical solutions across HR, finance, and operations modules
- Served as the primary technical advisor for key accounts, aligning delivery strategy with client objectives and long-term digital transformation roadmaps
- Established technical review and quality assurance processes, enforcing best practices in code standards, testing, and deployment procedures
- Collaborated with senior management to define product roadmaps, system enhancement pipelines, and strategic technical planning

IT Officer | iValue Management Consulting

Apr 2024 – Dec 2024

- Designed and developed custom Frappe applications within ERPNext, including automated cost distribution modules with integrated journal entry generation
- Engineered third-party integrations with ZATCA, Microsoft Teams, Zoom, and other enterprise platforms, extending ERPNext capabilities to meet regulatory and operational requirements

- Built and deployed interactive business intelligence dashboards connected to live ERPNext data sources, enabling real-time operational visibility for management
- Conducted comprehensive stakeholder workshops to capture business requirements, performed gap analysis, and produced detailed functional and technical specification documents
- Delivered end-user training programmes, authored user manuals and knowledge-base documentation, and provided post-go-live support across multiple client deployments
- Coordinated project planning and milestone tracking across multiple concurrent ERPNext implementations, ensuring alignment with agreed scope, budget, and delivery timelines

Full-Stack Developer | Wasac Management Consulting

Dec 2023 – Mar 2024

- Engineered custom ERP solutions from requirements through deployment, building scalable backend services, RESTful APIs, and responsive frontend interfaces using Python, C#, and JavaScript
- Facilitated stakeholder discovery sessions and translated business workflows into detailed technical specifications, user stories, and acceptance criteria
- Implemented security best practices, data protection measures, and conducted systematic code reviews to maintain quality and reliability standards across all deliverables

Full-Stack Developer | Quality Business Solutions (QBS)

Aug 2023 – Dec 2023

- Developed a feature-complete e-commerce platform encompassing product catalogue management, secure authentication, session-based shopping cart, and transaction processing
- Delivered three role-segmented dashboards — Admin, User, and Guest — with tailored functionality for inventory management, order tracking, account management, and guest checkout
- Optimised application performance and cross-device responsiveness, ensuring fast load times and a consistent user experience

TECHNICAL SKILLS

- **AI & ML Frameworks:** TensorFlow, Keras, Scikit-learn, PyTorch, LangFlow, OpenAI API, Hugging Face, YOLO (v8/v9), Roboflow, TensorBoard
- **Deep Learning:** Convolutional Neural Networks (CNN), Artificial Neural Networks (ANN), Generative Adversarial Networks (GAN), Transfer Learning (MobileNetV2, InceptionV3), RNNs
- **Machine Learning:** Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, SMOTE, Dimensionality Reduction (PCA)
- **NLP & RAG:** Retrieval-Augmented Generation (RAG), HyDE, RRF Reranking, Semantic Search, Vector Embeddings, TF-IDF, Sentiment Analysis, Text Preprocessing
- **Explainable AI:** SHAP Values, Permutation Importance, Model Interpretability
- **Data Science:** Pandas, NumPy, Matplotlib, Model Evaluation (F1-Score, mAP, Confusion Matrix, Precision-Recall)
- **Languages:** Python, JavaScript, TypeScript, C#, Java, C++, PHP
- **Web & Frameworks:** React, FastAPI, Django, ASP.NET Core MVC, HTML5, CSS3, Bootstrap, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, SQL Server, Oracle, SQLite
- **ERP:** ERPNext, Frappe Framework (custom DocTypes, workflows, server/client scripts, print formats, bench administration)
- **DevOps & Tools:** Docker, Git, Redis, Power BI, MS Project, Adobe XD, Microsoft 365

EDUCATION

BSc in Computer Information Systems | The Hashemite University

Oct 2019 – Oct 2023

ABET Accredited | GPA: Good

Senior Project: “Machine and deep learning classification for binary and multi-class network intrusion detection systems” — Published in IJAI, IAES Core (2025)

Publication: <https://ijai.iaescore.com/index.php/IJAI/article/view/27017/14858>

CERTIFICATIONS & TRAINING

- Machine Learning and Artificial Intelligence
- Deep Learning — Advanced
- Django Web Framework
- Microsoft SQL Server
- Oracle Database Administration
- Web Development — HTML, CSS, Bootstrap, JavaScript, PHP
- ASP.NET Core MVC Development

RESEARCH PUBLICATION

Hamdan, Y. (2025). “Machine and deep learning classification for binary and multi-class network intrusion detection systems.” *IJAI — Indonesian Journal of Electrical Engineering and Computer Science*, IAES Core.

DOI/Link: <https://ijai.iaescore.com/index.php/IJAI/article/view/27017/14858>

KEY PROJECTS

1. RYAI — Agentic AI Platform for ERPNext | iValue Management Consulting

Architected and delivered a production-ready Frappe application that introduces native agentic AI capabilities into ERPNext without external workflow dependencies. The core engine employs an OpenAI tool-use loop combining live SQL execution with semantic vector search (numpy), HyDE query expansion, RRF reranking, incremental indexing, and semantic caching. Additional modules include intelligent CV parsing from PDF and Word files via structured LLM calls, multi-LLM provider support (OpenAI, Google Gemini, Ollama), Redis-backed session persistence, permission-aware SQL generation, rate limiting, comprehensive audit logging, and field-level AI text enhancement injected across all ERPNext Long Text fields via SPA-compatible JavaScript.

Technologies: *Python, OpenAI API, Google Gemini, Ollama, Frappe/ERPNext, numpy, PyPDF2, python-docx, Redis, JavaScript*

2. SentriX — AI-Driven Security Operations Centre (SOC) Platform

Engineered a full-stack SOC platform from the ground up, providing real-time alert management with bulk actions, Kanban and list-view incident tracking, automated IOC enrichment via VirusTotal and AbuseIPDB APIs, a configurable alert-rule engine with auto-escalation, role-based user management, and multi-format report export (PDF, JSON, TXT). Integrated a RAG-based Virtual AI Analyst with live database context injection, persistent session history, and offline fallback mode. Developed a companion device monitor agent that detects suspicious processes, anomalous network activity, file integrity violations, SSH brute-force attempts, crontab modifications, and newly opened ports — streaming live alerts directly to the platform.

Technologies: *FastAPI, SQLAlchemy, SQLite, JWT, bcrypt, Python, psutil, Chart.js, VirusTotal API, AbuseIPDB API*

3. Website Builder & Content Management System

Developed a full-stack, drag-and-drop website builder featuring professionally designed templates across multiple industry categories. The backend (FastAPI + SQLAlchemy) implements a draft/publish workflow, page revision history, global layout management, media library, and a JWT-authenticated REST API. The React + GrapesJS frontend delivers a custom block library, extractable section panels, a theme customiser with colour presets and typography controls, and live preview with device emulation. An auto-responsive engine analyses page structure at runtime and injects appropriate CSS behaviours — grid collapse, flex-stack, fluid typography, and hamburger navigation — eliminating manual responsive work.

Technologies: *FastAPI, SQLAlchemy, React, GrapesJS, Zustand, TanStack Query, Tailwind CSS, TypeScript, Docker, SQLite*

4. Machine & Deep Learning for Network Intrusion Detection | Published Research

IJAI — Indonesian Journal of Electrical Engineering and Computer Science, IAES Core (2024)

Designed and evaluated machine learning and deep learning models — including Multilayer Perceptron, Random Forest, and Logistic Model Trees — for binary and multi-class network intrusion detection. Applied SMOTE for class imbalance correction and Pearson correlation for feature selection across NSL-KDD and CIC-IDS-2017 benchmark datasets.

Technologies: *Python, TensorFlow, Keras, Scikit-learn, Pandas, NumPy*

5. Explainable Multi-Class Intrusion Detection with GAN Architecture

Developed an advanced intrusion detection system comparing Multilayer Perceptron and novel Generative Adversarial Network architectures for multi-class attack classification on CIC-IDS 2017. Applied Explainable AI techniques — SHAP values and permutation importance — to surface model decision rationale and address interpretability challenges in cybersecurity deep learning deployments.

Technologies: *Python, TensorFlow, Keras, SHAP, Scikit-learn, Pandas, NumPy*

6. Automated Cost Distribution Financial System | iValue Management Consulting

Built an automated financial cost allocation engine within ERPNext that distributes payroll, employee benefits, and project costs across business units based on configurable contribution metrics and predefined partner ratios. The system auto-generates journal entries, manages cost reallocation from unprofitable to profitable projects, and delivers consolidated profit/loss tracking across projects and partnership structures.

Technologies: *ERPNext, Frappe Framework, Python, JavaScript*

7. Digital Transformation — Fertilizer Industry

Designed and implemented a comprehensive digitalisation programme for a fertilizer sector client, automating production monitoring, inventory management, and supply chain reporting. The solution introduced data-driven decision-making workflows, enhanced traceability across the production pipeline, and reduced operational inefficiencies through cross-functional collaboration with operations, logistics, and finance teams.

8. Computer Vision — Intelligent Seating Management System

Built a deep learning image classification system to automatically identify table occupancy and available seating capacity across real-world environments. Collected, annotated, and curated a custom dataset spanning five occupancy categories using transfer learning with MobileNetV2 and InceptionV3 to maximise accuracy on a constrained dataset.

Technologies: *Python, TensorFlow, Keras, MobileNetV2, InceptionV3, TensorFlow Hub, Transfer Learning*

9. Object Detection — University Furniture Recognition (YOLOv8 / YOLOv9)

Implemented a real-time object detection system to automatically identify and localise tables and chairs within university environments. Curated and augmented a custom annotated image dataset using Roboflow, training and benchmarking both YOLOv8 and YOLOv9 architectures with mAP-based evaluation.

Technologies: *Python, YOLOv8, YOLOv9, Roboflow, Computer Vision, Data Augmentation*

10. Obesity Pattern Analysis — Unsupervised Machine Learning

Conducted clustering analysis on obesity-related behavioural and physiological data, implementing K-Means and Hierarchical Clustering with PCA-based dimensionality reduction. Applied Elbow Method and Silhouette Analysis to validate optimal cluster configurations.

Technologies: *Python, Scikit-learn, K-Means, PCA, Hierarchical Clustering, Matplotlib*

11. Sentiment Analysis — NLP Pipeline for Recipe Reviews

Designed and executed an end-to-end NLP pipeline for multi-class sentiment classification of recipe review data. Implemented lemmatization, stop-word removal, and TF-IDF vectorisation, benchmarking Logistic Regression against Multinomial Naive Bayes classifiers.

Technologies: *Python, Scikit-learn, TF-IDF, Logistic Regression, Naive Bayes, Matplotlib*

12. Quatro-P4 — Full-Stack E-commerce Platform

Developed a full-featured e-commerce platform for pet care products with dynamic catalogue browsing, advanced search and filtering, session-managed shopping cart, and secure role-based authentication. Delivered dedicated dashboards for Admin (inventory and order management), registered Users (account and purchase history), and Guest (browsing and checkout).

Technologies: *HTML, JavaScript, PHP, MySQL*